

Refrigeration Simulation Report

Operating Conditions

Parameter	Value
compressor_efficiency	85.0
condenser_temp	45.0
evaporator_temp	-5.0
subcool	5.0
superheat	5.0

Performance

Metric	Value
COP	3.652
compressor_power	2.738
compressor_work	39.33
cooling_capacity	10.0
heat_rejected	182.97
mass_flow_rate	0.06962
refrigeration_effect	143.64
volumetric_efficiency	85.0

State Points

Point	P (kPa)	T (°C)	h (kJ/kg)	s	quality
1	243.34236987140005	0.0	400.0205270071958	1.7461483421951003	None
2	1159.924238342344	60.71429270369168	439.35200113240734	1.7639624262976212	None
3	1159.924238342344	40.0	256.3822763687963	1.1899916646448947	None
4	243.34236987140005	-4.99999999999943	256.3822763687963	1.210634183643984	0.31167979455454037